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CAMBRIAN PALEOMAGNETISM OF THE LLANO UPLIFT, TEXAS

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Abstract. Late Cambrian sandstones and limestones sampled from various members of the Riley and Wilberns formations of the Llano uplift show a progression of paleomagnetic pole positions as a function of age. The members, ages, and poles are the following for the Riley Formation: the Hickory Sandstone, Lower Dresbachian, 34°N, 145°E; the Cap Mountain Limestone, Dresbachian, 33°N, 140°E; the Lion Mountain Limestone, Upper Dresbachian, 24°N, 146°E. For the Wilberns Formation they are the following: the Welge Sandstone/Morgan Creek Limestone, Lower and Middle Franconian, 24°N, 151°E; and the Point peak Shale, Upper Franconian, 6°N, 159°E. These poles are based on thermal, chemical, and alternating field demagnetizations and on vector analysis. Most directions are interpreted to be of reversed polarity, but the Cap Mountain Limestone also yielded normal polarity directions; all directions resided in hematite with blocking temperatures up to 680°C. Almost all Cambrian poles appear to fall in a broad streak between the equator at about 155°E (e.g., the poles from the Tapeats Sandstone and from the Late Cambrian Point Peak Shale of the Wilberns Formation) and about 60°N, 90°E, where Cambrian poles have been obtained by Elston and Bressler (1977) and French et al. (1977). Although, at the present time, partial to complete remagnetization or nondipole behavior of the geomagnetic field are adequate ad hoc hypotheses to explain some of the data, it is suggested that the simplest and most unifying hypothesis to explain all data involves a Cambrian loop of apparent polar wander with respect to North America. This loop occurs before the middle Late Cambrian, with the poles from the Llano uplift falling on the return track.

Introduction

Paleomagnetic data from the lower Paleozoic rocks of North America are too limited to construct accurate paths of apparent polar wander. Recent publications [Gillett and Van Alstine, 1979; Van Alstine and Gillett, 1979; Watts and Van der Voo, 1979] reveal that significant portions of the path remain undocumented. Some of the more recent attempts at continental reconstructions [e.g., Morel and Irving, 1978; Kanasewich et al., 1978] embrace the premise that the classic Pangea restorations of Wegener [1929] or Du Toit [1937] have validity only for a short period of geological time and that large-scale displacements and plate reorganizations have occurred throughout the Phanerozoic. Without the benefit of seafloor anomalies, pre-Pangea reconstructions are more speculative than Mesozoic ones, and paleomagnetism offers the only hope of supplying quantitative data for Paleozoic and earlier time, albeit with certain limitations [e.g., Morel and Irving, 1978].

An important problem affecting Paleozoic continental restorations arises from the need to define the boundaries (margins) of continental blocks during that time. For a paleomagnetic result to be valid for an entire craton it must be demonstrated that the area sampled and the craton retained mutual coherence or that tectonic corrections can be applied that restore the sampled area to its ancient position with respect to this craton. In many instances, however, the validity of such corrections is questionable and, paradoxically, confirmed only by consistency with results from the craton (platform) itself. On the other hand, the thickest and most complete stratigraphic records are often from areas associated with deposition and tectonic activity at the margins. Platform deposits, though undeformed, are generally thin veneers of sediment with many gaps in the record. Also, because of the relatively undisturbed setting of platform deposits the fold test of Graham [1949] cannot always be applied with any great degree of statistical significance. To establish the relative validity of the probable ages of a magnetization, one must sometimes appeal to a combination of indirect arguments involving stratigraphic position, the presence of normal and reversed polarities that are antiparallel, the internal consistency of results from various parts of the continental shield, or relative coercivities and blocking temperatures of various magnetic components.

The importance of paleomagnetic data from platform deposits is that tectonic rotations may be discounted a priori. Since such rotations affect paleomagnetically determined declinations and since previously published Cambrian results suggest declination variations of 70° or more [French et al., 1977; Elston and Bressler, 1977], the Cambrian sedimentary rocks of the Llano uplift of central Texas are of interest in the documentation of the lower Paleozoic segment of the North American apparent polar wander path.

Geological Setting of the Llano Uplift of Texas

Figure 1 is a generalized map of the Llano uplift. Paleozoic sedimentary formations are ex-

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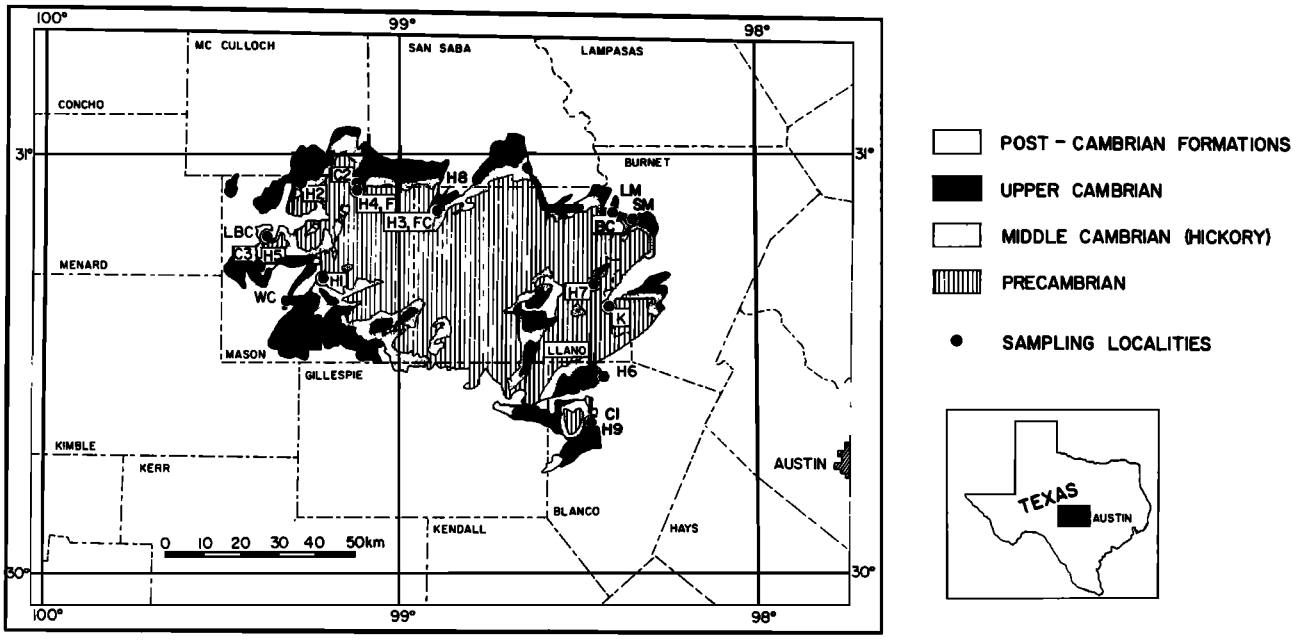


Fig. 1. Generalized geologic map of the Llano uplift of Texas, indicating paleomagnetic sampling sites as given in Table 1.

posed around the edges of a Precambrian crystalline core. The present structural uplift was generated by late Paleozoic movements along boundary faults [Bell and Barnes, 1972].

The Cambrian stratigraphy of the Llano uplift is summarized in Figure 2 [after Bell and Barnes, 1972]. For a more complete description of the stratigraphy the reader is referred to that work and to Barnes and Bell [1977]. The Hickory member of the Riley Formation represents a basal transgressive sequence, which overlies the Precambrian crystalline basement with a profound unconformity. Late Middle Cambrian (Albertan) trilobite species of Bolaspidella are found low in this unit, whereas the sampled upper portions of the Hickory are Dresbachian in age. The sampled strata consisted of hematitic sandstones which have potential for economic exploitation [Barnes and Schofield, 1964]. The origin and age of the hematite is problematic, since it coats detrital grains and fills interstices between grain boundaries. Biogenetic fragments appear to have been entirely replaced by hematite so at least a portion of the hematite is not detrital. It has been suggested by Barnes and Schofield [1964] that the hematite was formed from iron contained in iron oxy-hydroxides (goethite, limonite, etc.) which was later transformed into hematite upon burial. Replacement of chamosite and glauconite may also have been responsible.

Conformably overlying the Hickory in a gradational contact is the Cap Mountain Limestone Member of the Riley Formation. The Cap Mountain contains hematitic sandstones in its lower beds and grades upward into a glauconitic limestone. This unit ranges in thickness from 50 to 165 m.

Conformably overlying the Cap Mountain is the Lion Mountain Sandstone Member, stratigraphically the highest member of the Riley Formation. It consists of glauconitic quartz-sandstones 10 to 23 m in total thickness. As can be seen in Figure 2, the contact between the Riley and Wilberns formations is a disconformity, reflecting a hiatus in the sedimentary record due to a marine regression at the Dresbachian-Franconian boundary according to Bell and Barnes [1972]. The Welge Member of the Wilberns Formation is a coarse to medium grained nonglauconitic sandstone, 3 to 12 m thick and representing a transgression of the Franconian sea.

Conformably overlying the Welge Member is the Morgan Creek Limestone Member of the Wilberns Formation. It is uniformly 43 to 47 m thick over the entire Llano uplift and consists of fossiliferous glauconitic limestones representing deposition in a shallow Franconian sea.

The Point Peak Shale Member of the Wilberns Formation conformably overlies the Morgan Creek Limestone and consists of shale, siltstone, conglomerate, stromatolitic bioherms, and biostromes with a total thickness ranging from 8 to 72 m. It is followed by the San Saba Member of the Wilberns Formation. Deposition of this unit spanned Trempealeauan time and continued into the Ordovician on the western side of the Llano uplift.

Sampling and Laboratory Treatment

Paleomagnetic sampling of the Cambrian of the Llano uplift was undertaken independently by two research groups, one from the University of Michigan (D. R. Watts and R. Van der Voo) and the other from the University of Texas at Dallas (S. C. Reeve).

Both groups used a portable coring apparatus, and orientations were measured using a clinometer and a Brunton compass. The number of samples collected from each unit was as follows: 165 oriented cores from 10 sites in the Hickory Member, 48 cores from 3 sites in the Cap Mountain Member, 51 cores from 2 sites in the Lion Mountain Member, 88 cores from 2 sites in the Morgan Creek-Welge Members, and 24 cores from 1 site in

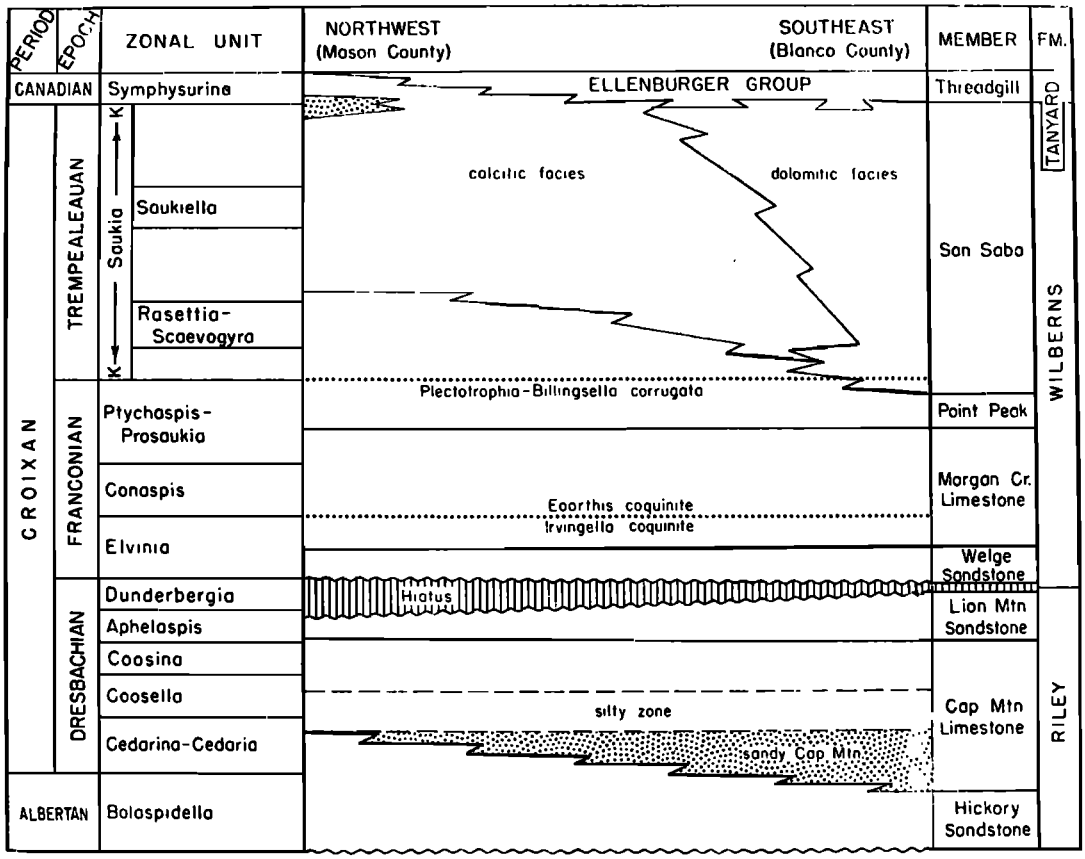


Fig. 2. Generalized stratigraphic column of the Cambrian sediments of the Llano uplift [from Bell and Barnes, 1972].

the Point Peak Shale of the Wilberns Formation (earlier results for this member were reported by Van der Voo et al. [1976]). The San Saba Member was also sampled, but these specimens were not demagnetized and will not be discussed here. Site locations are shown in Figure 1.

Samples were measured in both laboratories with a cryogenic magnetometer. Stepwise demagnetization procedures were employed, and the results were plotted in orthogonal projections [Zijderveld, 1967]. At the University of Michigan, treatment was carried out until the final intensity was 10 to 1% of the total natural remanent magnetization (NRM) or until a stable endpoint was revealed. It was found that alternating field (AF) demagnetization results in incomplete removal of secondary components. Chemical leaching was only partially successful because of the high carbonate content and the coarse to medium grained nature of many of the samples. Many samples disintegrated during the first 10 hours of leaching in 8N HCℓ. The most successful method of isolating characteristic directions was thermal demagnetization, which resulted in the most linear trajectories for the vectors, as revealed by the Zijderveld diagrams.

The samples collected by the Dallas group were thermally treated with typically three and at most four steps of demagnetization between 275°C and 610°C. AF demagnetizations were performed on selected samples in a series of six steps. Apparently secondary components were still present after maximum demagnetizing fields of about 70 mT had been applied. At this point, diagrams were constructed to assess the degree of demagnetization, but no attempt was made to obtain characteristic directions by vector subtraction. At a later stage in the investigation the data from the University of Texas at Dallas were replotted in new Zijderveld diagrams so that a valid comparison could be made between the two sets of data from the two laboratories, and vector subtractions were performed in order to obtain characteristic directions as revealed by three or more collinear points. For stable endpoints the origin of the diagram was considered as a point. In many cases, it could be demonstrated that the level of demagnetization and number of steps applied were insufficient to resolve all of the components in a sample and that further treatment should have been applied. Such results were not used in the calculation of a mean direction. Only those results that were interpretable in terms of characteristic directions were used to calculate the mean directions.

Results: Hickory Member

Figure 3a shows the stable behavior exhibited by a number of specimens from the Hickory Member in response to thermal demagnetization, i.e., a uniform, linear decay of the magnetization vector expressed by three or more collinear points. Figure 3b depicts results from a chemical leaching of a sample of similar lithology, again revealing a component of magnetization character-

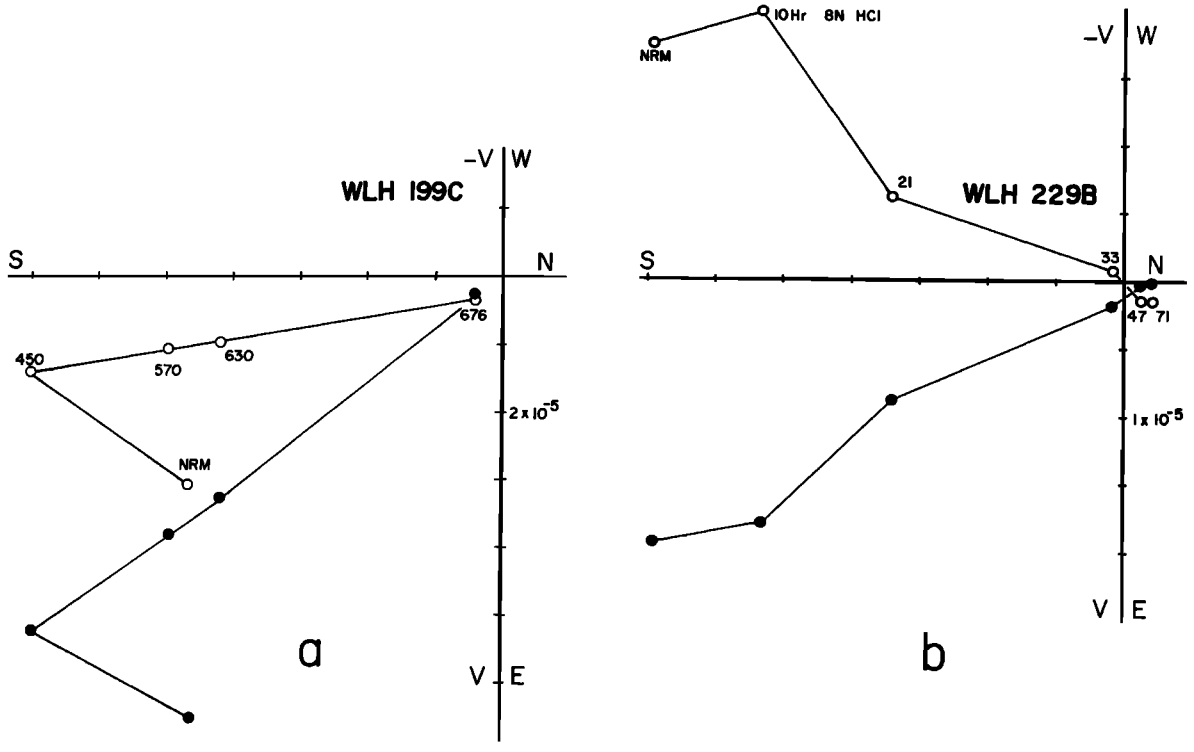


Fig. 3. Orthogonal projections [Zijderveld, 1967] of progressive demagnetizations of samples from the Hickory Member (Riley Formation). Open circles are projections onto the vertical plane and solid circles are projections onto the horizontal plane. Units indicated for the axes are in electromagnetic units per cubic centimeter (10^{-3} emu/cm^3 = 1 A m^{-1}). (a) Thermal and (b) chemical demagnetization; cumulative leaching time is indicated in hours.

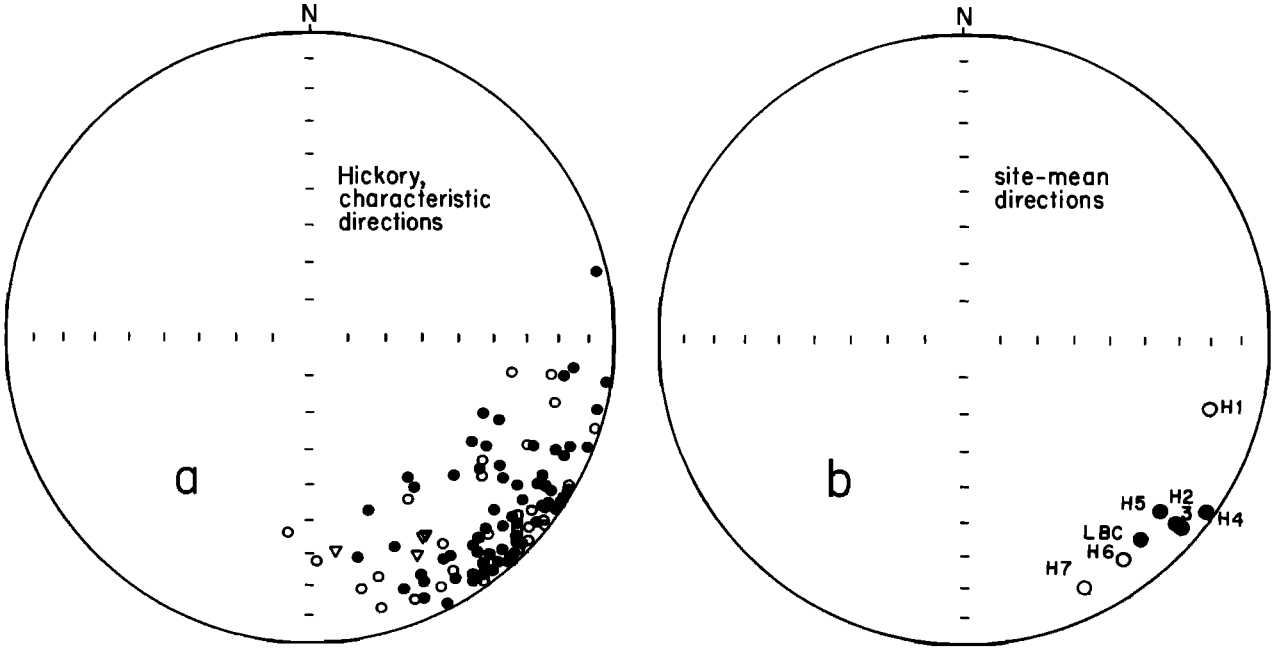


Fig. 4. Equal area projection of (a) the characteristic directions of magnetization and (b) site mean directions, after correction for the tilt of the strata, for the Hickory Member of the Riley Formation. Open circles denote upper-hemisphere projections, solid circles denote lower-hemisphere projections; circles (triangles) are directions obtained from thermal (chemical) demagnetization.

TABLE 1. Site Mean Directions, Statistics, and Pole Positions

Site	Location, deg		R_s	Before Tilt Correction			After Tilt Correction				Pole Position		dp	dm
				Decl	Incl	k	Decl	Incl	k	α_{95}				
Hickory Member: Characteristic Site Means														
Site H1	30.72°N,	99.22°W	4/6	106°	-13°	11	103°	-17°	11	29°	16°N,	172°E	15°	30°
Site H2	30.95°N,	99.29°W	11/14	130°	+9°	23	130°	+9°	23	10°	31°N,	143°E	5°	10°
Site H3-FC	30.85°N,	98.90°W	48/54	134°	+7°	19	134°	+7°	19	5°	34°N,	141°E	2°	5°
Site H4-F	30.93°N,	99.12°W	15/18	126°	+15°	15	126°	+1°	14	10°	30°N,	151°E	5°	10°
Site H5	30.76°N,	99.40°W	8/10	131°	+11°	32	130°	+16°	35	10°	29°N,	140°E	5°	10°
Site H6	30.47°N,	98.43°W	10/15	147°	0°	6	146°	-12°	6	21°	50°N,	141°E	11°	21°
Sity H7	30.76°N,	98.40°W	10/12	155°	+17°	11	154°	-12°	12	14°	56°N,	131°E	7°	15°
Site H8	30.85°N,	98.90°W	0/9	---	---	--	---	---	--	--	-----	-----	--	--
Site H9	30.37°N,	98.46°W	0/4	---	---	--	---	---	--	--	-----	-----	--	--
Site LBC	30.82°N,	99.38°W	6/23	143°	+16°	8	143°	+16°	8	25°	37°N,	129°E	13°	25°
Cap Mountain Member: Characteristic Site Means														
Site Cl_n	30.37°N,	98.46°W	9/9	299°	+12°	20	300°	+12°	21	12°	29°N,	161°E	6°	12°
Site Cl_r	30.37°N,	98.46°W	11/11	148°	+6°	44	148°	+6°	44	7°	45°N,	130°E	4°	7°
Site C2	30.93°N,	99.12°W	15/15	129°	+36°	6	120°	+30°	6	17°	16°N,	141°E	10°	19°
Site C3	30.76°N,	99.40°W	11/13	145°	+4°	33	145°	+10°	35	8°	41°N,	130°E	4°	8°
Lion Mountain Member: Characteristic Site Means														
Site LM	30.85°N,	98.35°W	13/36	141°	+9°	18	140°	+10°	18	10°	38°N,	135°E	5°	10°
Site K	30.65°N,	98.42°W	6/15	112°	+13°	20	112°	+13°	20	15°	15°N,	154°E	8°	16°
Morgan Creek-Welge Members: Characteristic Site Means														
Site SM	30.85°N,	98.40°W	25/29	120°	+9°	17	120°	+9°	17	7°	23°N,	151°E	4°	7°
Site BC	30.87°N,	98.40°W	40/59	121°	-9°	23	121°	+6°	23	5°	25°N,	151°E	2°	5°
Point Peak Member: Characteristic Site Mean														
Site WC	30.65°N,	99.32°W	15/24	97°	+3°	18	97°	+3°	18	9°	6°N,	166°E	5°	9°

Mean	R_s	Before Tilt Correction			After Tilt Correction				Pole Position		dp	dm
		Decl	Incl	k	Decl	Incl	k	α_{95}				
Hickory, mean of all samples	112/165	134°	+4°	11	135°	+9°	12	4°	34°N,	140°E	2°	4°
Hickory, mean of the poles	8/10			28			23	12°	36°N,	145°E		
Cap Mountain, mean of samples	46/48	136°	+13°	8	134°	+11°	8	8°	33°N,	140°E	4°	8°
Cap Mountain, mean of poles	4/4			28			28	15°	33°N,	141°E		
Lion Mountain, mean of samples	19/51	132°	+10°	[12]	132°	+10°	[13]	[10°]	32°N,	142°E	[5°]	[10°]
Lion Mountain, mean of poles	2/2			--			--	--	27°N,	146°E		
Morgan Creek-Welge, sample mean	65/88	121°	-2°	16	121°	+7°	20	4°	24°N,	151°E	2°	4°
Morgan Creek-Welge, pole mean	2/2			--			--	--	24°N,	151°E		

Site numbers correspond to those of Figure 1; Decl and Incl are declination and inclination, respectively, in degrees; R_s is the ratio of samples or sites used to derive a particular mean direction or pole to the total number of samples or sites analyzed; k is the estimate of Fisher's precision parameter for a population of paleomagnetic directions or poles; and α_{95} is the associated radius of the cone of 95% confidence [McElhinny, 1973]; dp and dm are the semiaxes of the oval of 95% confidence about the poles calculated from mean paleomagnetic directions; the statistical parameters are entered in brackets where distributions were non-Fisherian.

istic of the Hickory Member. Not all samples behaved as those depicted in Figure 3. Some had linear trajectories but did not decay uniformly toward the origin, indicating the presence of unresolved high-temperature magnetic components. Other samples had distinctly nonlinear trajectories (randomwalk) on the diagrams, and results from these were classified as uninterpretable and were not used to calculate mean directions.

The directions characteristic of the Hickory Member are plotted in equal area projection in Figure 4a, with site mean directions in Figure 4b and listed in Table 1. All of these components were revealed by linear trajectories on Zijderveld diagrams. The blocking temperatures of these components typically ranged between 550°C and 675°C.

Results: Cap Mountain Member

Orthogonal projections of the demagnetization of three samples are given in Figure 5. Results of thermal (Figure 5a) and chemical (Figure 5b) demagnetization of a buff calcareous siltstone sample taken from the middle portion of the Cap Mountain Limestone (above the stippled zone; in-

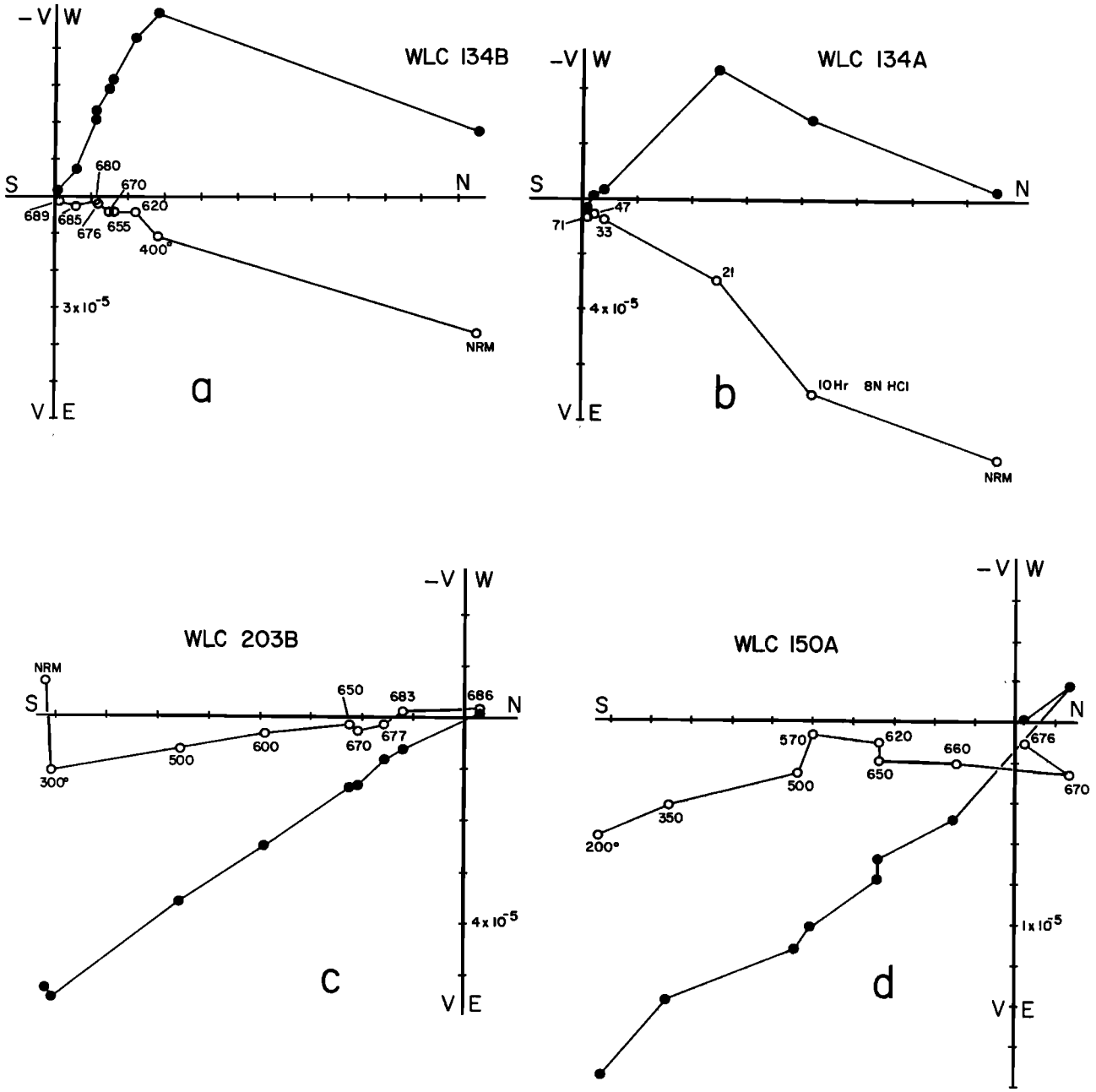


Fig. 5. Demagnetization results (as in Figure 3) of samples from the Cap Mountain Limestone Member of the Riley Formation. (a) Thermal and (b) chemical demagnetization of specimens from the same sample with northwesterly (normal polarity) directions of magnetization; (c) sample with reversed polarity magnetization and (d) sample with both reversed and normal polarity components of magnetization.

dicated as silty in Figure 2) at site C1 show west-northwesterly declinations which are interpreted as being of normal polarity. Figures 5c-5d show demagnetizations of red hematitic rocks at sites C3 and C1 from the lowest beds of the Cap Mountain Member, with a predominantly reversed polarity. Figure 5d, however, indicates that in this sample both normal and reversed polarity components are present, with the reversed component apparently overprinting the normal (northwesterly) component; at least one of the components must be younger (secondary) than the age of the beds. In all Cap Mountain samples (e.g., Figures 5a-5d) the blocking temperatures indicate that hematite is the carrier of the characteristic magnetization. Unfortunately, at this site sampling was not carried out with magnetostratigraphic results in mind; outcrops sampled were not continuous, moreover.

Figure 6a shows the characteristic directions of the silty and sandy parts of the Cap Mountain Limestone Member. The site mean directions are given in Table 1 and plotted in Figure 6b. In the calculation of the mean directions for site C1, normal polarity directions were averaged separately from reversed polarity directions. From the buff siltstones at site C2, 3 directions of normal polarity and 12 of reversed polarity were isolated and averaged together (Table 1) into a reversed polarity mean. Figure 6b shows that the site mean directions are not precisely antipodal. Site C3 is in the highest portion of the Streeter section [Barnes and Schofield, 1964] only a few meters above the extensively sampled section of the Hickory Member (site H5) and probably stratigraphically below site C2. Site C1 has not been directly correlated with C2 and C3 but appears to span the same interval of time in the section, so it is at present not quite clear what causes the directions to be less than 180° apart. Discussion of the significance of normal polarity components with northwest declinations in North American Cambrian rocks is deferred to the interpretation section.

Results: Lion Mountain Member

Thermal demagnetization results from the Lion Mountain Sandstone are given in Figure 7, representative of the two sites sampled. Directions of these samples were originally chosen [Reeve, 1975] on the basis of minimum contamination by present-day field overprinting; later the measurements were reanalyzed as previously described, using Zijderveld diagrams. Although many of the demagnetizations appeared to be incomplete (e.g., Figure 7b), a number of samples behaved as depicted in Figure 7a, allowing the isolation of characteristic directions from a significant number of demagnetizations. These characteristic directions are plotted in Figure 8a, and the site mean directions are given in Table 1 and Figure 8b. There is considerably more variation in declination than in inclination when the two sites are combined (as in Figure 8a), and the distribution of all sample directions does not warrant the application of Fisher's statistics [McElhinny, 1973].

Results: Welge/Morgan Creek Members

The samples from the Welge Sandstone and the Morgan Creek Limestone members of the Wilberns Formation were collected from two stratigraphically overlapping sections with a continuous exposure of both members in order to evaluate their potential for magnetostratigraphy. All directions were found to be of reversed polarity, however, and the results of both sec-

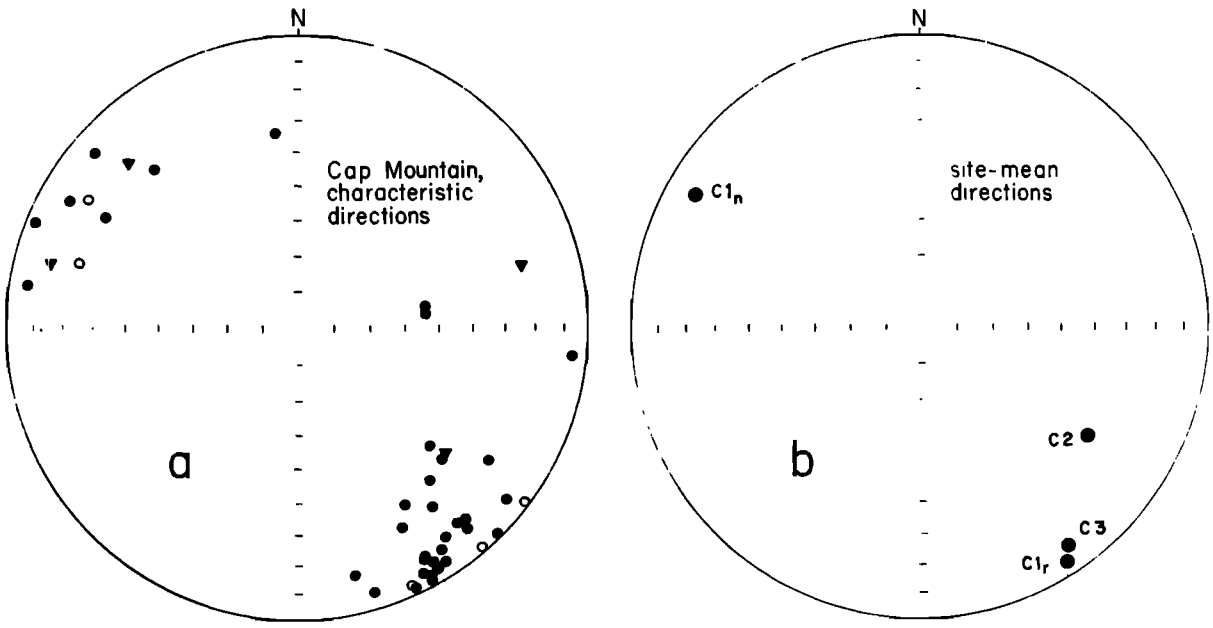


Fig. 6. Equal area projection of (a) the characteristic directions and (b) the site-mean directions for the Cap Mountain Limestone Member of the Riley Formation. The normal and reversed polarity samples were averaged separately for site C1, whereas 3 normal polarity samples of site C2 were included in the total average for that site. Symbols are as defined in Figure 4.

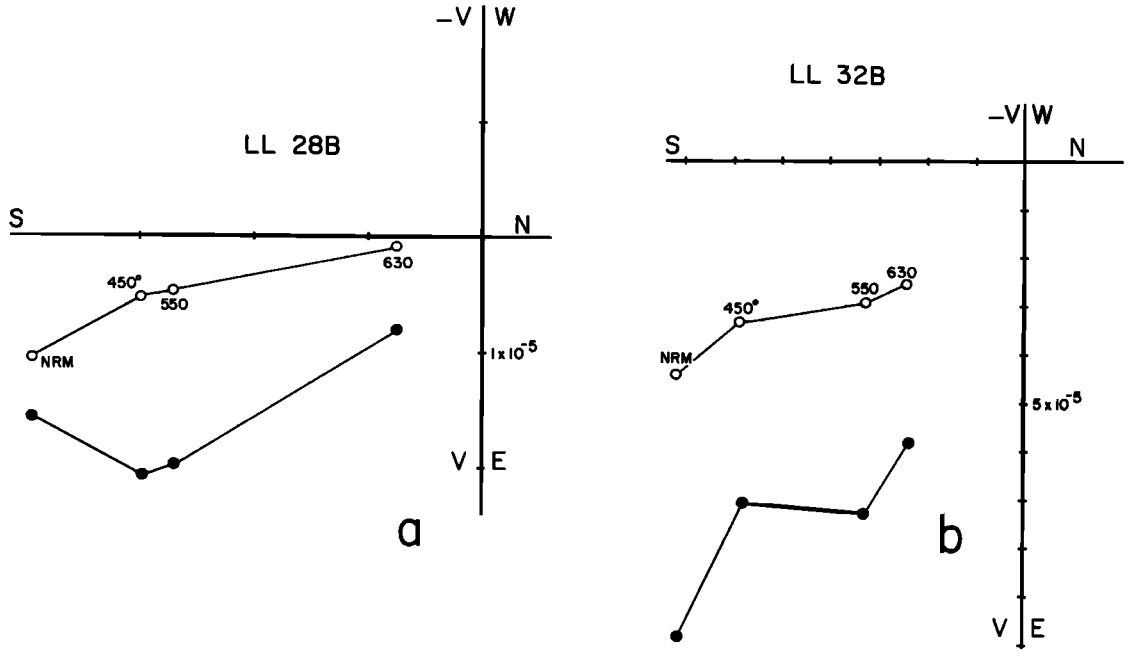


Fig. 7. Demagnetization results (as in Figure 3) of samples from the Lion Mountain Sandstone Member of the Riley Formation. (a) A well-defined characteristic component with at least three collinear points and (b) a typical incomplete demagnetization of a sample not used in the calculation of the final site mean directions.

tions are averaged here together for the determination of a pole position. Figure 9a shows the demagnetization of a sample from the Welge Sandstone, Figure 9b from the Morgan Creek Limestone. Initial selection [Reeve, 1975] was identical to that of the Lion Mountain Member. The Zijderveld diagrams constructed for the present analysis show that it was possible to isolate characteristic components from a number of samples. These characteristic directions are plotted in Figure 10a, and Figure 10b gives the sitemean directions (see also Table 1). The equal area projection of the sample directions (Figure 10a) shows similar magnitudes of variation in inclination and declination (both about 60°), and the distribution is close to Fisherian.

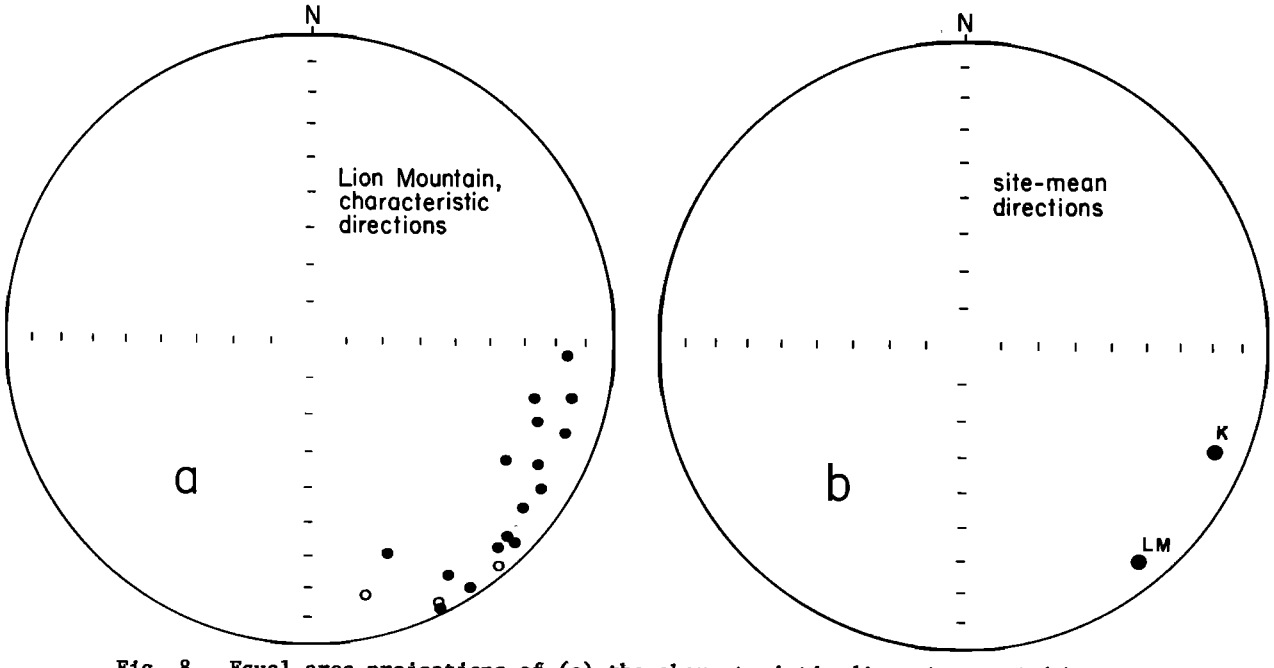


Fig. 8. Equal area projections of (a) the characteristic directions and (b) the site mean directions, after corrections for the tilt of the strata, for the Lion Mountain Sandstone Member of the Riley Formation. Symbols are as in Figure 4.

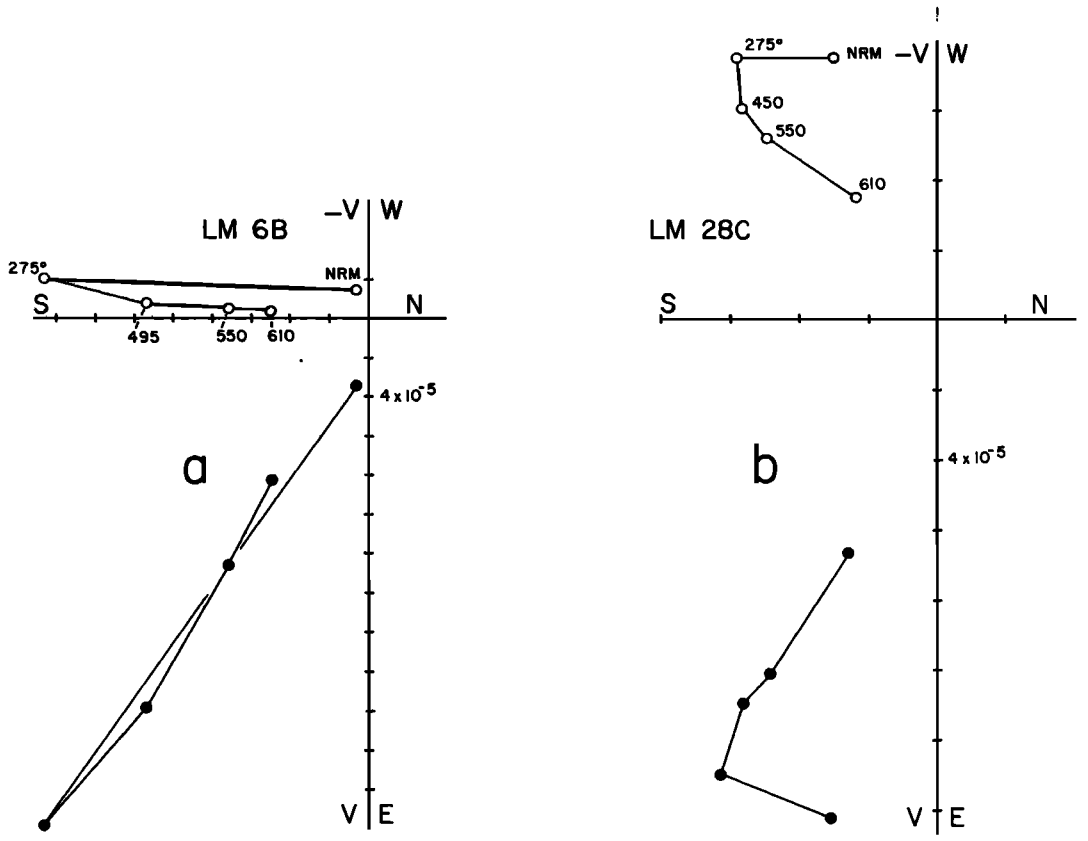


Fig. 9. Demagnetization results (as in Figure 3) of samples from (a) the Morgan Creek Limestone Member and (b) the Welge Sandstone Member of the Wilberns Formation.

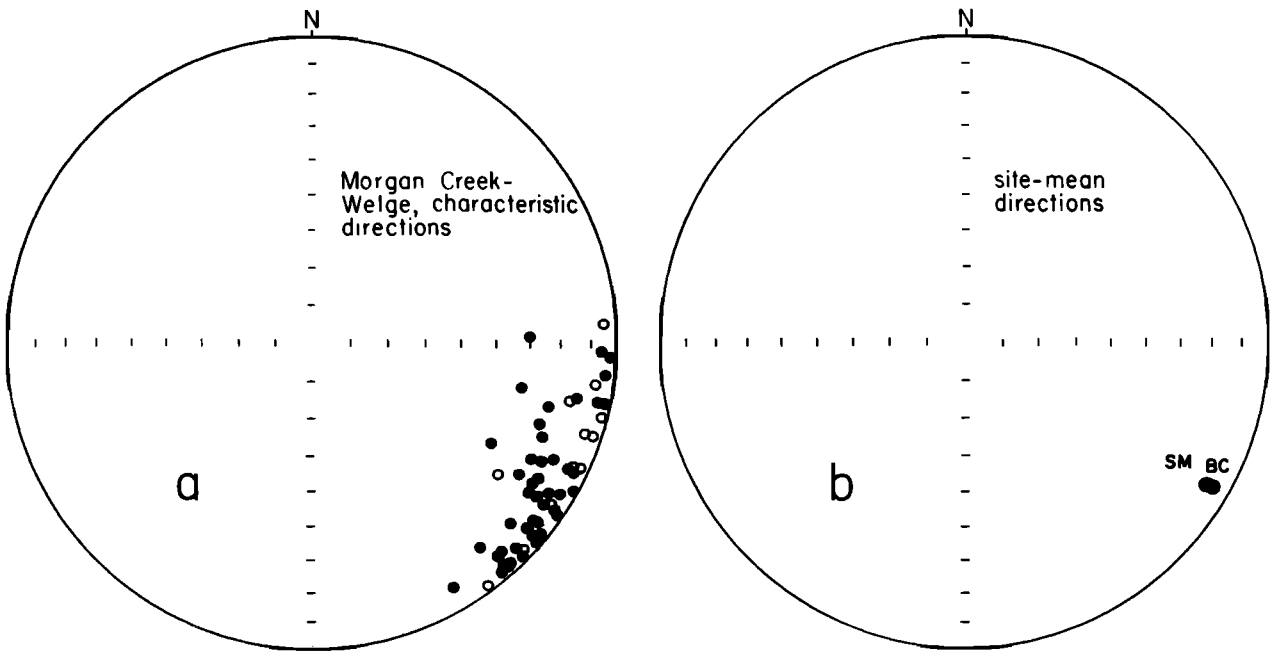


Fig. 10. Equal area projections of (a) the characteristic directions and (b) the site mean directions, after correction for the tilt of the strata, for the Morgan Creek and Welge members of the Wilberns Formation. The results from these members were indistinguishable and have been averaged together. Symbols are as defined in Figure 4.

Results: Point Peak and San Saba Members

One site in the Point Peak Shale Member of the Wilberns Formation gave a pole at 6°N, 166°E that confirms the results reported earlier by Van der Voo et al. [1976]; it is important to note that the new site is at the opposite side of the Llano uplift and in rocks with different lithology and structural correction from those sampled by Van der Voo et al. Their pole position for the Point Peak Member is located at 6°N, 159°E. From the San Saba Member of the Wilberns Formation, also, one site was sampled, but the samples were not demagnetized because of obvious secondary carbonate and iron mineral diagenesis.

Discussion of Results

The mean poles obtained from the various members of the Riley and Wilberns formations are plotted in Figure 11. The mean Cap Mountain and Hickory poles fall very close to one another, which is not surprising in view of the conformable contact between these units with the Cap Mountain samples coming from the lowest portions of this member. The Hickory and Cap Mountain poles, the Lion Mountain pole, the Welge-Morgan Creek pole, and the Point Peak pole form a regular progression with the poles from the oldest rocks falling the closest to upper Paleozoic poles for cratonic North America. This progression of poles is reflected by a continuous decrease in declination with decreasing age. The Hickory pole is separated from the Point Peak pole by about 30° of arc.

A large majority of the samples contained magnetizations which clearly resided in hematite on the basis of blocking temperatures. These magnetizations are therefore very likely to be chemical remanent magnetizations. Most paleomagnetic workers assume that red beds and other hematitic sediments acquire their magnetizations within at most a few million years after deposition, but the possibility of a much later age of the magnetization cannot a priori be excluded from consideration. Consequently, a large part of the following section will be devoted to the arguments for and against such a later (re)magnetization.

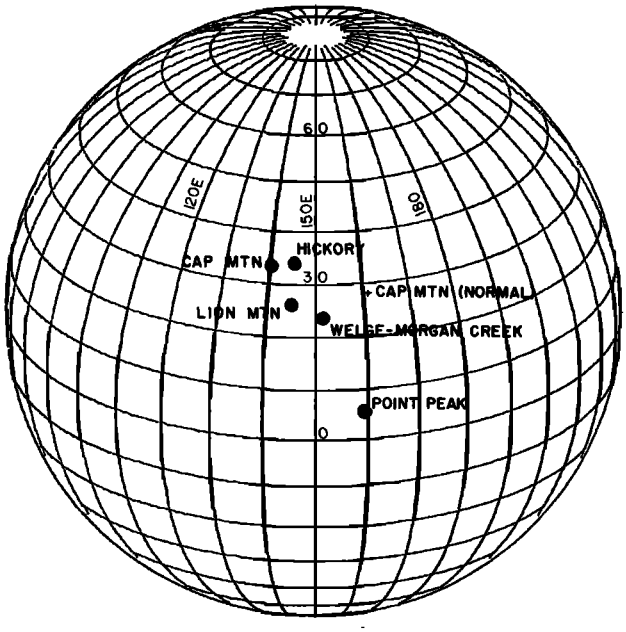


Fig. 11. Mean pole positions for the Cambrian units of the Llano uplift of Texas. The pole determined from the C1 site with normal polarity directions is plotted as a cross. Note the progressive southeastward migration of the pole with younger age.

Interpretation

The Cambrian has been problematic in the documentation of the North American path of apparent polar wander. Various workers [Al-Khafaji and Vincenz, 1971; Larson and Mutschler, 1971; Vincenz et al., 1975; French, 1976; French et al., 1977; Elston and Bressler, 1977; Patterson et al., 1978; Gillett and Van Alstine, 1979; Dankers and Lapointe, 1979] have reported results from Cambrian rocks of North America which as an ensemble show a dispersal in declinations and pole positions streaked between the equator at about 160°E and the cluster of late Paleozoic poles for North America (at about 45°N, 120°E or beyond). All Cambrian paleomagnetic poles interpreted as representative of the North American craton are listed in Table 2a and plotted in Figure 12. The remagnetization hypothesis is often invoked to explain this distribution of poles [Larson and Mutschler, 1971; Vincenz et al., 1975], as are tectonic rotations in some instances [Gillett and Van Alstine, 1979]. The possibility of a loop of apparent polar wander has been examined by Van der Voo [1978].

A difficulty in interpreting Cambrian paleomagnetic results is the determination of the time of acquisition of the magnetic components. Results from igneous rocks can be dated on the basis of the time of closure of various isotopic systems, but the analytical errors often preclude the determination of relative ages of magnetization [e.g., French et al., 1977]. From sedimentary strata the relative ages may be inferred from stratigraphic position and from application of the fold test where possible, but 'absolute' ages remain unknown or indirectly determined. Moreover, the age of magnetization may not be the same as the age of deposition but rather the age of diagenesis or alteration of the iron-bearing phases sometime during the history of the rock.

Figure 12 shows pole positions derived from both igneous and sedimentary rocks of latest Precambrian and Cambrian age. These poles come from cratonic areas for which tectonic rotations (about local vertical axes) are generally very unlikely. A number of interpretations of the dispersion of these results is possible.

1. The trend represents partial to complete late Paleozoic or early Mesozoic remagnetization; the intermediate poles result from the vector addition of a Cambrian component and a late Paleozoic to early Mesozoic component. Aspects of this remagnetization scheme were first hypothesized by Creer [1968].

2. The trend is the result of a series of remagnetization events from Ordovician through

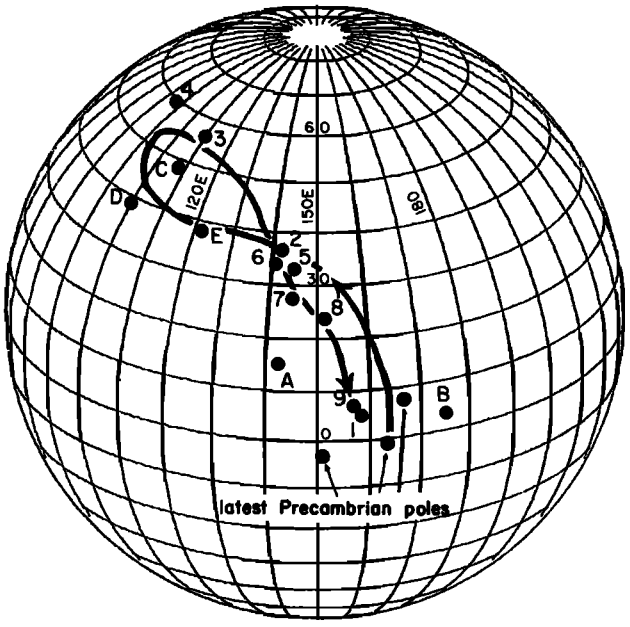


Fig. 12. Positions of latest Precambrian and Cambrian paleomagnetic poles representative of the North American craton (Table 2a). Cambrian poles derived from sedimentary units are numbered and those from igneous units are lettered in presumed stratigraphic order. The diagram shows the progressive shift of the Cambrian apparent polar wander path from the latest Precambrian poles to the Middle Cambrian Abrigo pole (4) and from there back to the Late Cambrian Point Peak Shale pole (9). Pole D is the only well-dated result (525 Ma) from the igneous rocks for which coordinates are published.

Triassic times, with most Cambrian aged rocks undergoing complete remagnetization.

3. The trend reflects nondipole behavior of the Cambrian geomagnetic field, manifested by large swings in declination.

4. The trend is the result of apparent polar wander with respect to North America during the Cambrian.

We will consider the results from the Llano uplift of Texas in the context of these possibilities. The southeasterly declinations of the members older than the Point Peak Shale could be considered as the vector sum of a southerly late Paleozoic/early Mesozoic component and an easterly Cambrian component [Reeve, 1975]. However, the normal polarity directions (with northwest declinations) observed in the Cap Mountain Member in this study argue against this interpretation. If these magnetizations were the vector sum of a normal polarity Cambrian component and a southerly late Paleozoic/early Mesozoic component, the declinations would be drawn toward the southwest not the northwest. A secondary component with northerly declinations remains a possibility, but it would be fortuitous if normal polarity Cambrian components would be overprinted by normal polarity late Paleozoic/early Mesozoic components, while all reversed Cambrian components are overprinted by reversed secondary components. In addition, the late Paleozoic has an overwhelming dominance of reversed (Kiaman) directions. Furthermore, the components in this study were isolated from linear demagnetization trajectories as revealed by Zijderveld diagrams; an interpretation of our directions in terms of only partial remagnetization forces us to assume that the blocking temperature spectra of the Cambrian and secondary components in all cases are quite identical. It is unlikely that this is the case, and our chemical demagnetizations revealed that the solubility spectra show no bimodality either. We feel we can safely exclude a partial remagnetization as an explanation for the directions of this study.

Complete remagnetization during Ordovician or later time cannot be discounted. Such an event would give rise to normal and reversed polarities with linear demagnetization trajectories and high blocking temperatures or coercivities. However, it is considered quite unlikely that the remagnetization would occur in the Cambrian and Carboniferous [Kent and Opdyke, 1979] rocks in such a manner that it would result in the regular progression of poles with age as revealed by Figure 11, complemented by the early Carboniferous pole of Kent and Opdyke [1979] at 49.1°N, 119.3°E for the Barnett Formation of the Llano uplift. While our magnetization directions reside almost exclusively in hematite, Kent and Opdyke found no indications of hematite as a carrier of their Carboniferous magnetizations. We conclude that if complete remagnetizations must be held responsible for our set of poles, then these remagnetizations occurred at different times for different members and, presumably, through different mechanisms.

Nondipole behavior of the Cambrian geomagnetic field, resulting in large-scale (60°) swings in magnetic declination, would explain the results from the Llano uplift. However, the concept of normal and reversed polarities has meaning only in the context of a dipole field. It is considered unlikely that nondipole behavior would give rise to anomalous normal and reversed directions of magnetization, as observed in this study and by French et al [1977], Dankers and Lapointe [1979], and Watts et al [1980]. Cambrian paleomagnetic results from other continents have not given rise to suspicions about nondipole field behavior either [McElhinny and Embleton, 1976; Hagstrum et al., 1980].

Since all results listed in Table 2a or plotted in Figure 12 are derived from rocks of the craton (except for the Rome Formation), relative rotations of the sampling sites may be discounted. Consequently, we contend that the simplest and most unifying hypothesis to explain the presently available Cambrian poles is that of apparent polar wander, i.e., a North American continental rotation with respect to the geomagnetic pole. As will be explained below, this rotation probably occurred during Middle and early Late Cambrian time, with the Llano poles falling on the return track of a Cambrian apparent polar wander loop. The validity of this hypothesis can only be tested by the internal consistency of results from other parts of the craton as it existed during the Cambrian or from orogenic belts in which the appropriate structural corrections are known with some certainty.

TABLE 2a. Latest Precambrian and Cambrian Paleomagnetic Poles Representative of the North American Craton

Number	Formation	Age	Pole Position		Reference
9	Wilberns, Point Peak Member	Upper Franconian	6°N,	159°E	Van der Voo et al. [1976]
8	Wilberns, Morgan Creek-Welge members	Lower to Middle Franconian	24°N,	151°E	this paper
7	Riley, Lion Mountain Member	Upper Dresbachian	27°N,	146°E	this paper
6	Riley, Cap Mountain Member	Lower to Middle Dresbachian	33°N,	140°E	this paper
5	Riley, Hickory Member	Lower Dresbachian	34°N,	145°E	this paper
4	Abrigo Formation	Middle to Upper Cambrian	59°N,	89°E	Elston and Bressler [1977]
3	Muav Formation	Middle Cambrian	55°N,	110°E	Elston and Bressler [1977]
2	Rome formation	Upper-Lower Cambrian	38°N,	142°E	Watts et al. [1980]
1	Tapeats Formation	Lower Cambrian	5°N,	158°E	Elston and Bressler [1977]
E	Colorado intrusives	Cambro-Ordovician	37°N,	122°E	Larson and Mutschler [1971]
D	Colorado intrusives	525 m.y.	39°N,	100°E	Patterson et al. [1978]
C	Colorado intrusives, group III	upper Precambrian-Lower Ordovician	48°N,	107°E	French et al. [1977]
B	Colorado intrusives, group II	upper Precambrian-Lower Ordovician	5°N,	174°E	French et al. [1977]
A	Colorado intrusives, group I	upper Precambrian-Lower Ordovician	15°N,	142°E	French et al. [1977]
-	Grenville dikes	uppermost Precambrian	3°S,	151°E	Murthy [1971]
-	Coronation sills	647 m.y.	1°S,	163°E	Robertson and Baragar [1972]
-	Franklin diabases	675 m.y.	8°N,	167°E	Fahrig et al. [1971]

TABLE 2b. Cambrian Poles With Problematic Age of Magnetization or Tectonic Control

Formation	Age	Pole Position		Reference
Bradore Formation, Newfoundland	Lower Cambrian	29°N,	167°E	Rao and Deutsch [1976]
Jangle and Desert Limestone members, Carrara Formation; Papoose Lake Member, Bonanza King Formation; uncorrected for rotation	Middle Cambrian	37°N,	136°E	Gillett and Van Alstine [1979]
Same, corrected for rotation of 36°	Middle Cambrian	9°N,	160°E	
Gold Ace, Red pass, and Pahrump members, Carrara Formation; uncorrected	Lower and Middle Cambrian	36°N,	134°E	Gillett and Van Alstine [1979]
Same, corrected for rotation of 36°	Lower and Middle Cambrian	7°N,	158°E	
Wood Canyon Formation; uncorrected	Lower and Middle Cambrian	28°N,	134°E	Gillett and Van Alstine [1979]
Wood Canyon Formation; corrected for rotation	Lower and Middle Cambrian	1°S,	157°E	
Ophiolite Complex, Quebec	550 m.y.	13°N,	146°E	Seguin [1976]
Lamotte Formation, thermal demagnetization	Upper Cambrian	1°N,	168°E	Al-Khafaji and Vincenz [1971]
Bonneterre Dolomite	Upper Cambrian	35°S,	190°E	Beales et al. [1974]
Wichita Granites	525 m.y.	29°N,	142°E	Ku et al. [1967]
Wichita Granites, AF demagnetization	525 m.y.	13°N,	147°E	Spall [1970]
Wichita Granites, thermal demagnetization	525 m.y.	4°N,	164°E	Spall [1970]
Wichita Granites, AF demagnetization	525 m.y.	16°N,	148°E	Vincenz et al. [1975]
Wichita Granites, thermal demagnetization	525 m.y.	30°N,	148°E	Vincenz et al. [1975]

Examination of the distribution of poles (Figure 12) shows a systematic variation with age: the poles from sedimentary rocks are numbered according to stratigraphic position beginning with the Lower Cambrian Tapeats pole as number 1. Other results from the North American craton include data also from the Colorado Plateau, i.e., the Muav and Abrigo formations [Elston and Bressler, 1977], which give well-defined poles near Late Devonian, Permian, and Triassic poles for North America, approximately 70° of arc away from the Tapeats and Late Cambrian Point Peak Shale poles (1 and 9, respectively, in Figure 12). Since these units were sampled near a Cambrian-Devonian disconformity, the possibility of remagnetization due to near-surface processes associated with groundwater movement during the Devonian has been suggested by Elston and Bressler [1977]. However, igneous rocks from Colorado have given poles near the Muav and Abrigo poles, and it is from these that the most convincing refutation of the possibility of remagnetization can be obtained. The Colorado igneous rocks have been studied by a number of workers. Larson and Mutschler [1971] isolated components that yielded pole E of Figure 12. French et al. [1977], in a more extensive survey, found three distinct sets of components, yielding poles A, B, and C which deviate by various amounts from the Point Peak and Tapeats poles. At the time these results were obtained, conclusive and precise age determinations were not available for the rocks sampled. More recently Armbrustmacher et al. [1979] have obtained Rb/Sr ages of 530 to 510 Ma for syenites and nepheline syenites in the alkaline complexes studied by French et al. [1979]. Patterson et al. [1978], moreover, have obtained pole D in Figure 12 from intrusives dated at 525 Ma. All workers [Larson and Mutschler, 1971; French et al., 1977; Patterson et al., 1978] argued that the rocks studied were very fresh and that there were no reasons to believe that any remagnetizations had occurred after the acquisition of the original thermoremanence.

The available stratigraphic and radiometric ages suggest that the poles numbered 2-8, or lettered C, D, and E are predominantly of Middle and early Late Cambrian age, or about 540 to 525 Ma according to the ages reported for the igneous rocks. It seems clear that the Point Peak Shale pole is younger than these poles; stratigraphically its age is Upper Franconian, whereas a radiometric age determination of glauconites with Rb-Sr methods [Morton and Long, 1978] gives 507 ± 12 Ma.

It appears therefore that apparent polar wander can explain the presently available Middle and Late Cambrian poles, if we assume a trend that begins with poles 3, 4, C, D, and E and ends with the Point Peak Shale pole, while passing during late Middle and early Late Cambrian time through the other poles of Figure 12. In a companion paper, Watts et al. [1980] have presented a discussion of the latest Precambrian and the Early to Middle Cambrian poles for North America. Their conclusion was that the apparent polar wander path for that time moved over about 70° of arc from the equator at 165°E (latest Precambrian) to the group of poles near 50°N, 100°E. They also noted that the only result which does not fit the temporal aspects of this trend is the Tapeats pole and speculated that perhaps the Tapeats sandstone was remagnetized in Late Cambrian times. Combining the conclusions of this paper with those of the companion paper, we arrive at a loop in the Cambrian apparent polar wander path. Such a loop involves a back-and-forth rotation of the American continent with respect to the geomagnetic pole. After the Late Cambrian the pole once more swings to the northwest in the western Pacific to arrive at the Middle Ordovician poles obtained from the Moccasin and Bays formations at 33°N, 147°E [Watts and Van der Voo, 1979].

Results for Cambrian rocks with problematic age, paleomagnetic, or tectonic control are listed in Table 2 b and can for the most part be explained very well by the same Cambrian loop of apparent polar wander path (APWP). Only some of the results of the Lamotte Formation [Al-Khafaji and Vincenz, 1971] and the Bonneterre Dolomite [Beales et al., 1974] appear to fall outside the proposed trend. We note, however, that these results were fairly scattered and that demagnetization experiments in some instances may have been incomplete; AF demagnetizations have been reported that were apparently inadequate to resolve the potentially multivectorial nature of the NRM in many Lamotte samples.

In summary, late Precambrian and Cambrian poles with reasonably well-determined ages and good structural control (Table 2a) progress from the latest Precambrian poles to the Rome pole and on to the Middle Cambrian Muav and Abrigo poles. From lower Dresbachian to Upper Franconian time the pole moves progressively back to the Point Peak pole. The loop's radiometrically well-dated result (pole D, from Patterson et al. [1978]) has an age of 525 Ma, which according to the time scale of Van Eysinga [1975] corresponds well to Middle Cambrian. It seems very unlikely that remagnetization played a role in the remanence acquisition of the intrusive rocks from Colorado, and equally unlikely that remagnetization processes would respect stratigraphic age in the sedimentary units. For some of the poles, Ordovician or younger complete remagnetizations cannot be entirely ruled out, but this would not explain all available poles. We propose therefore that a Cambrian APWP loop occurred and that the poles from the Llano uplift represent the return path of this loop. Granted a coaxial, geocentric Cambrian dipole field, this loop would be due to an initial counterclockwise rotation, followed by a clockwise rotation of the North American craton during the late Middle and early Late Cambrian, with respect to the pole.

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